

Tianqi Xu

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Google Scholar | LinkedIn | GitHub

EDUCATION

Sichuan University (Project 985)

Bachelor of Science in Software Engineering

- GPA: 3.77 / 4.0; Weighted Average: 89.32 / 100
- Advisor: Tao He

Chengdu, China

Sept. 2023 – June 2027 (Expected)

RESEARCH INTERESTS

Target Directions: Embodied AI, World Models.

Current Focus: Test-Time Adaptation, Ordinary Differential Equations, Medical Image Segmentation.

RESEARCH EXPERIENCE

University of Central Florida

Remote Research Intern; Advisor: Yuzhang Shang

Mar. 2026 – Present

Orlando, FL

- Investigating the integration of Chain-of-Thought (CoT) reasoning mechanisms into Vision-Language-Action (VLA) models to enhance embodied reasoning and long-horizon capabilities.
- Developing a self-supervised Reinforcement Learning (RL) strategy for the post-training phase of VLA models, aiming to optimize policy alignment and improve sequential decision-making without extensive manual annotations.

Sichuan University

Research Assistant; Advisor: Tao He

Aug. 2025 – Feb. 2026

Chengdu, China

- Proposed SaCTTA, a Continual Test-Time Adaptation method to mitigate domain shifts in medical image segmentation.
- Designed an attention-driven dynamic layer selection mechanism that treats layer outputs as information tokens to adaptively select discriminative layers and enhance inter-layer interaction.
- Introduced a self-distilled adaptive learning rate with dual constraints (sensitivity & consistency) to accelerate adaptation and suppress error accumulation.
- Co-first author paper submitted to NeurIPS 2026.**

Sichuan University

Research Assistant; Advisor: Tao He

Sep. 2024 – July 2025

Chengdu, China

- Proposed CNM-UNet, a medical image segmentation model that replaces vanilla hierarchical decoders with a single Continuous Neural Memory ODEs Block to minimize computational complexity.
- Leveraged Neural Memory ODEs for continuous temporal feature extraction, enabling precise data distribution modeling and enhanced accuracy.
- Designed a Dual Self-updating strategy based on test-time adaptation principles to enhance cross-domain generalization.
- First author paper presented in AAAI 2026.**

PUBLICATION

Tianqi Xu, Yashi Zhu, Quansong He, Yue Cao, Kaishen Wang, Zhang Yi, Tao He. CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation, **AAAI 2026**. (Website & Paper & Code)

HONORS AND AWARDS

- Merit Student Award (Consecutive 2 Years)**, Sichuan University 2023 – 2025
- Second-class Comprehensive Academic Scholarship (**Top 6%**), Sichuan University 2023 – 2024
- Third-class Comprehensive Academic Scholarship (**Top 15%**), Sichuan University 2024 – 2025

SKILLS

Languages: English (CET-4: 585, CET-6: 560), Chinese (Native)

Programming: Python, C/C++, JAVA, SQL

Others: PyTorch, Linux, Git, Docker, OpenCV, $\text{\LaTeX}$